

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	N.SRIKANTH	Reg. No.:	192411139
Section / Batch:	CSE	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I N.SRIKANTH(192411139) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Q1 Sports Tournament Ranking System (Quick Sort)

100 marks

A sports management system ranks teams based on scores and performance metrics. Analyze how Quick Sort can be applied for dynamic ranking updates. Identify issues such as pivot selection and worst-case scenarios. Apply divide and conquer concepts to explain its working. Analyze efficiency and propose improvements with justification.

Your Code

```
1 #include <stdio.h>
2
3 void swap(int *a, int *b)
4 {
5     int temp = *a;
6     *a = *b;
7     *b = temp;
8 }
9
10 int partition(int arr[], int low, int high)
11 {
12     int pivot = arr[high];
13     int i = low - 1;
14     int j;
15
16     for(j = low; j < high; j++)
17     {
```

Input

```
Enter number of teams: 5
Enter team scores:
78 95 66 88 91
```

Output

```
Enter number of teams: Enter team
scores:

Team Ranking (High to Low):
Rank 1 : 7143516 points
Rank 2 : 6881377 points
```

Aim

To implement Quick Sort algorithm for a sports management system that ranks teams based on their scores and performance metrics in descending order.

Algorithm: Quick Sort for Team Ranking

1. Start.
2. Read the list of team scores.
3. Select a pivot element.
4. Partition the list:
 1. Place teams with higher scores before the pivot.
 2. Place teams with lower scores after the pivot.
5. Recursively apply Quick Sort on left and right partitions.
6. Display the sorted team rankings.
7. Stop.

Python Code

```
# Quick Sort for Sports Team Ranking

def quick_sort(teams):

    if len(teams) <= 1:

        return teams

    pivot = teams[len(teams) // 2]

    left = [x for x in teams if x > pivot]

    middle = [x for x in teams if x == pivot]

    right = [x for x in teams if x < pivot]

    return quick_sort(left) + middle + quick_sort(right)

# Team scores

teams = [85, 70, 95, 60, 80]

print("Original Team Scores:")

print(teams)

# Sorting teams based on performance

rankings = quick_sort(teams)

print("\nTeam Rankings (Highest to Lowest):")

print(rankings)
```

Conclusion:

Quick Sort successfully ranks teams based on their scores using the divide-and-conquer approach with an average time complexity of $O(n \log n)$.

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				

Total Marks (100):	
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Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____